International Journal of Forecasting 42 (2026) 1123–1134

Contents lists available at ScienceDirect

International Journal of Forecasting

journal homepage: www.elsevier.com/locate/ijforecast

Does the consideration of market prices in model selection increase model profitability? Evidence from theory, artificial data and real-world data

Fabian Wunderlich *, Marc Garnica Caparrós, Daniel Memmert

Institute of Exercise Training and Sport Informatics, German Sport University Cologne, Am Sportpark Müngersdorf 6, 50933, Cologne, Germany

ARTICLE INFO

Article history:
Dataset link: https://github.com/spoho-datascience/market-model-profitability

Keywords:
Profitable forecasting
Economic value
Accuracy
Model selection
Artificial data
Sports forecasting

ABSTRACT

It is well-acknowledged in the literature that besides accuracy, profitability (i.e. the economic value) of a forecast is crucial in domains including economics, energy, weather or sports forecasting. The present study argues that selection of profitable forecasting models benefits from consideration of market prices in-sample (i.e. when optimizing model parameters on training data). Such models can be specifically optimised for profitability instead of accuracy, as most commonly done so far. Previous literature suggests that, under specific circumstances, accuracy- and profitability-optimisation leads to equal model selection. We theoretically show that this result is contingent upon strict assumptions and does not hold anymore if relaxing any of these assumptions. Moreover, we present evidence that out-of-sample profitability (i.e. returns generated on unseen data) are actually increased for profitability-optimised models compared to accuracy-optimised models based on artificial and real-world data from the domain of sports betting. This result is explained by the ability of profitability-optimised models to implicitly detect and counter an existing market bias, provided that the model family allows to capture such biases. The present study intends to stimulate discussion on optimal model selection for profitable forecasting and to encourage researchers to consider profitability-optimisation in forecasting applications to real-world data.

© 2025 The Authors. Published by Elsevier B.V. on behalf of International Institute of Forecasters. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

1. Introduction

Forecasting-related decision making not only requires expertise, availability of information, careful consideration, and sophisticated methods, but can also crucially be influenced by the goal of the forecast. Without knowledge of the forecasting goal, it may not even be clear what a "good" forecast is. The best-researched forecasting goals in the literature are arguably accuracy and profitability (i.e., economic value), and the ambiguous relationship between these two goals in forecasting has been discussed for decades (Boothe & Glassman, 1987; Fuertes, Kalotychou, & Todorovic, 2015; Hobbs, Jitprapaikulsarn, Konda, Chankong, Loparo, & Maratukulam, 1999; Hong, Pinson, Wang, Weron, Yang, & Zareipour, 2020; Leitch & Tanner, 1991; Murphy, 1993; Zhu, Toth, Wobus, Richardson, & Mylne, 2002).

The relevance of profitability is most apparent in forecasts directly linked to economic indicators such as stock prices (Ertimur, Sunder, & Sunder, 2007), exchange rates (Boothe & Glassman, 1987), or interest rates (Leitch & Tanner, 1991). But further forecasting domains are also

* Corresponding author.
E-mail addresses: f.wunderlich@dshs-koeln.de (F. Wunderlich), m.garnica@dshs-koeln.de (M. Garnica Caparrós), d.memmert@dshs-koeln.de (D. Memmert).

https://doi.org/10.1016/j.ijforecast.2025.11.006
0169-2070/© 2025 The Authors. Published by Elsevier B.V. on behalf of International Institute of Forecasters. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

embedded in strongly economic contexts. Energy forecasting, for example, implies financial consequences for energy supply companies or energy-intensive industries, which means that there can be an "economic value of reducing forecast error" (Hong et al., 2020, p. 385). Even in the domain of weather forecasting, which does not appear to be primarily economically motivated, forecasts may evoke economic consequences such as wins or loses of a farmer due to weather-dependent decisions. Zhu et al. (2002) even state that "the ultimate measure of the utility of weather forecasts is arguably the economic and other benefits associated with their actual use in the daily decision-making process of individuals and different organizations" (p. 74).

It may be appealing to assume that the most accurate models always lead to the highest economic value, and under specific circumstances, such consistency of goals is even backed up by theory (Johnstone, 2011). Researchers from various domains of forecasting, however, have presented results that cast doubt that forecasting accuracy and profitability always coincide. Fuertes et al. (2015) state that their analysis of stock market volatility "reinforces the view that statistical significance does not have a direct mapping onto economic value" (p. 251) when statistical loss functions and volatility-based trading rules are compared. Similarly, it has been found that the most accurate exchange rate forecasting models are not necessarily the most profitable models in forward market speculation (Boothe & Glassman, 1987). Leitch and Tanner (1991) even argue that models not being able to outperform naïve benchmarks in terms of statistical accuracy can have financial value. Moreover, in an analysis of interest rate forecasts, they find that forecasting errors "have no systematic relationship to profits" (p. 580). In energy forecasting, it has been reported that the economic impact of imperfect electricity market price forecasts on operation planning in industry depends on the exact demand of the market participant (Zareipour, Canizares, & Bhattacharya, 2009); that the relationship between accuracy and economic value is not linear and not monotone in the area of solar or wind forecasts translating into power forecasts (Yang & Kleissl, 2022); and that economic costs are typically increased for higher forecasting error (i.e., lower accuracy), but also depend on the direction of error in electric load forecasting, where underforecasts are more costly than overforecasts (Hobbs et al., 1999). In weather forecasting, it has been reported that optimal decisions based on ensemble (probabilistic) forecasts are dependent on the individual cost loss ratio of the users (Zhu et al., 2002) and that risk aversion affects decisions of farmers based on forecasts (Cerdá Tena & Quiroga Gómez, 2011).

Despite these apparent differences between accuracy and profitability, forecasting models intended to generate economic value are often selected by optimizing for accuracy. The present research analyzes whether, in contrast to this common approach, it can be beneficial to consider the goal of forecasting in the model selection process. We do this by comparing the economic value of profitability-optimized models already measuring economic consequences (by including market prices) in model selection with that of accuracy-optimized models considering only observed outcomes in model selection. Conclusions are based on theoretical considerations as well as analysis of artificial and real-world data.

For the latter, we utilize data from the domain of sports forecasting (Wunderlich & Memmert, 2021), which offers several convenient characteristics helping both to motivate and to answer our research question. First, for sports forecasts there exists a natural representation of economic value, namely returns in the sports betting market, where model profitability is highly crucial to confirm the financial relevance of a forecasting model (Constantinou, Fenton, & Neil, 2012; Koopman & Lit, 2015) or present evidence for violations (Angelini & Angelis, 2019; Bernardo, Ruberti, & Verona, 2019; Direr, 2011) of efficient market hypotheses (Fama, 1970). Second, high availability and low restrictions for data in professional sports provide us with the possibility to gather and analyze a large dataset of real-world data including both event results and market prices (i.e., betting odds). Third, consistent with other domains, there is previous work suggesting that financial success in the market (through identification of profitable betting strategies) does not necessarily require an accurate forecasting model (Hubáček & Šír, 2023; Wunderlich & Memmert, 2020). These results are based on the idea that (even inaccurate) models performing the opposing forecasting error than the market can be used to exploit market inaccuracies. Particularly, profitability increases if the correlation between market and forecaster error decreases (Wunderlich & Memmert, 2020), and consequently models may increase profitability by being decorrelated with the market (Hubáček & Šír, 2023; Hubáček, Šourek, & Železný, 2019). Fourth, despite these results, it appears to be state-of-the art in sports forecasting to optimize models for accuracy even if analyzing or explicitly aiming at model profitability (Boshnakov, Kharrat, & McHale, 2017; Hvattum & Arntzen, 2010; Koopman & Lit, 2015, 2019).

2. Theoretical evidence

The idea that forecasting goals should be considered in model selection is supported by Gneiting (2011), who argues that effective forecasts require ex ante knowledge of the scoring function, at least for point forecasts. However, Johnstone (2011) provides evidence that models chosen by maximum likelihood also optimize Kelly betting returns, which, in contrast, suggests independence of model selection from the goal of the forecast. We show that the result of Johnstone (2011) only holds under the very specific assumptions made in the formal proof, of which three are absence of margins (i.e., the market is commission-free), multiplicative Kelly betting (i.e., the stake for Kelly betting is determined as a fraction of the current wealth), and only two possible outcomes of an event. Each of these assumptions might easily be violated in real-world situations, and we show that relaxing any of these assumptions is enough to reach a different conclusion. To do so, no formal proof is needed; it is sufficient to construct example models, where the consistency of optimized likelihood (accuracy) and optimized Kelly betting returns (profitability) does not hold.

We consider a series of $n = 100$ events with each having m possible outcomes (e.g., tennis matches with $m = 2$ or football matches with $m = 3$). For reasons

Table 1
Example models where the more accurate model (higher likelihood) does not correspond to the more profitable model (higher wealth after betting) when relaxing three assumption of the proof of Johnstone (2011). Asterisks refer to the higher likelihood or wealth after betting respectively.

Assumption relaxed	Absence of margins		Multiplicative Kelly betting		Only two possible outcomes	
True Probabilities	$(p_1, p_2) = (0.6, 0.4)$		$(p_1, p_2) = (0.6, 0.4)$		$(p_1, p_2, p_3) = (0.50, 0.25, 0.25)$	
Market Probabilities	$(\bar{p}_1, \bar{p}_2) = (0.5, 0.5)$		$(\bar{p}_1, \bar{p}_2) = (0.5, 0.5)$		$(\bar{p}_1, \bar{p}_2, \bar{p}_3) = (0.25, 0.25, 0.50)$	
Betting Odds	$(o_1, o_2) = (1.8, 1.8)$		$(o_1, o_2) = (2.0, 2.0)$		$(o_1, o_2, o_3) = (4.0, 4.0, 2.0)$	
Model	Accurate	Profitable	Accurate	Profitable	Accurate	Profitable
Model probabilities	$(\tilde{p}_1, \tilde{p}_2) = (0.65, 0.35)$	$(\tilde{p}_1, \tilde{p}_2) = (0.50, 0.50)$	$(\tilde{p}_1, \tilde{p}_2) = (0.50, 0.50)$	$(\tilde{p}_1, \tilde{p}_2) = (0.80, 0.20)$	$(\tilde{p}_1, \tilde{p}_2, \tilde{p}_3) = (0.25, 0.25, 0.50)$	$(\tilde{p}_1, \tilde{p}_2, \tilde{p}_3) = (0.70, 0.05, 0.25)$
Model stakes	$(s_1, s_2) = (0.2125, 0.0)$	$(s_1, s_2) = (0.0, 0.0)$	$(s_1, s_2) = (0.0, 0.0)$	$(s_1, s_2) = (0.60, 0.0)$	$(s_1, s_2, s_3) = (0.0, 0.0, 0.0)$	$(s_1, s_2, s_3) = (0.60, 0.0, 0.0)$
Log-Likelihood	$\mathbf{-29.46}^*$	-30.10	$\mathbf{-30.10}^*$	-33.77	$\mathbf{-52.68}^*$	-55.32
Betting returns	-0.13	$\mathbf{0.00}^*$	0.00	$\mathbf{12.00}^*$	0.00	$\mathbf{288.00}^*$

of simplicity, we assume that the outcomes of events are independent and that the probability of a specific event outcome $i = 1 \dots m$ to occur, denoted as p_i, is equal across all events. The outcome probabilities are estimated (more or less correctly) by both the market ($\bar{p}_i$) and the forecasting model ($\tilde{p}_i$). The betting odds o_i are the inverse of the market probabilities, but slightly reduced in the case of existing margins. The stakes in a Kelly betting strategy are calculated as $s_i = \frac{\tilde{p}_i \cdot o_i - 1}{o_i - 1}$. We construct probabilities in such a way that $100 \cdot p_i$ is an integer value, and thus we are able to assume that the observed fraction of each outcome exactly equals its probability (i.e., $100 \cdot p_i$ occurrences), which implies a log-likelihood of $\log(\prod_{i=1}^{m} \tilde{p}_i^{100 \cdot p_i})$. In a multiplicative Kelly strategy, the wealth is reduced by s_i in every bet and increased by $s_i o_i$ if the event actually occurs; thus the betting returns are calculated as $\prod_{i=1}^{m} [(1 + s_i(o_i - 1))^{100 \cdot p_i} \cdot (1 - s_i)^{(100 \cdot (1 - p_i))}] - 1$. If a multiplicative strategy is not used, the stake is considered a fixed amount (i.e., wins and losses are not reinvested) and thus the betting returns are calculated as $\sum_{i=1}^{m} [100 \cdot p_i \cdot s_i (o_i - 1) - 100 \cdot (1 - p_i) \cdot s_i]$. Table 1 shows example models violating the consistency of accuracy and profitability when each of the three assumptions is relaxed separately. Please note that, as evidenced by Johnstone (2011), we would not have been able to construct such models if all assumptions were fulfilled. In the absence of margins, the increased betting odds on outcome 1 would have made the accurate model profitable. In multiplicative Kelly, investigating the high amount of 60% of wealth invested in the profitable model would have been penalized by multiplicative betting, which does not occur when a fixed amount of wealth is invested. Moreover, for only two outcomes, we could not have constructed the blatant underestimation of p_2 in the profitable model that leads to a disadvantageous likelihood.

3. Application to data

The theoretical evidence is an important prerequisite, but examples are highly constructed and not sufficient to imply transferability of results to more realistic and complex situations. Thus, we develop a framework applicable to more complex datasets grounded in the domain of sports betting (where we intend applicability to both artificial datasets generated by simulation and real-world datasets). The framework is summarized in Table 2 and—in analogy to the theoretical considerations—contrasts the ground truth (i.e., the outcome probabilities), the market/bookmaker forecast used as the basis to determine market prices, and the model/bettor forecasts that can be used in an attempt to place profitable bets in the market. Testing the in-sample selection of a forecasting model requires us to first choose a model (or model family), whose parameters require optimization. Please note that in the artificial data we will use the very same model family to represent our simulated ground truth on outcome probabilities, as well as the estimates performed by bookmaker and forecaster. The bettor then optimizes across several possible model parameters and selects the model from the model family having the best accuracy or profitability in-sample, respectively. We follow the approach of Hvattum and Arntzen (2010) by proposing an ordered logistic regression (subsequently OLR) model with a single covariate, namely the difference of team ratings $\Delta = r_H - r_A$, where r_H is the current rating of the home team and r_A is the current rating of the away team. The match outcome probabilities for home win p_H, draw p_D, and away win p_A in this logistic model are given by

$$p_H(\Delta) = 1 - \frac{1}{1 + \exp(-c_1 + \beta\Delta)}$$

$$p_D(\Delta) = \frac{1}{1 + \exp(-c_1 + \beta\Delta)} - \frac{1}{1 + \exp(-c_0 + \beta\Delta)}$$

$$p_A(\Delta) = \frac{1}{1 + \exp(-c_0 + \beta\Delta)}.$$

Accordingly, three parameters, c_0, c_1, and β, are required to fully specify an instance of this probability model. When referring to a *model* subsequently, we do not refer to the mathematical model, but to a specific instance (i.e., model specification). Please refer to Fig. 1 for an illustration of the model for parameters $c_0 = -0.9$, $c_1 = 0.3$, and $\beta = 0.006$ (the specification we will use later in the artificial data). The x-axis shows the difference between the rating points of home and away team, Δ, i.e., the difference in team strength. The y-axis represents the probability of the three outcomes of a football match.

4. Evidence from artificial data

4.1. Data generation

A dataset of 152,000 matches is simulated, based on generating the schedule of play for 40 leagues consisting of 20 teams in a double round-robin system (i.e., being

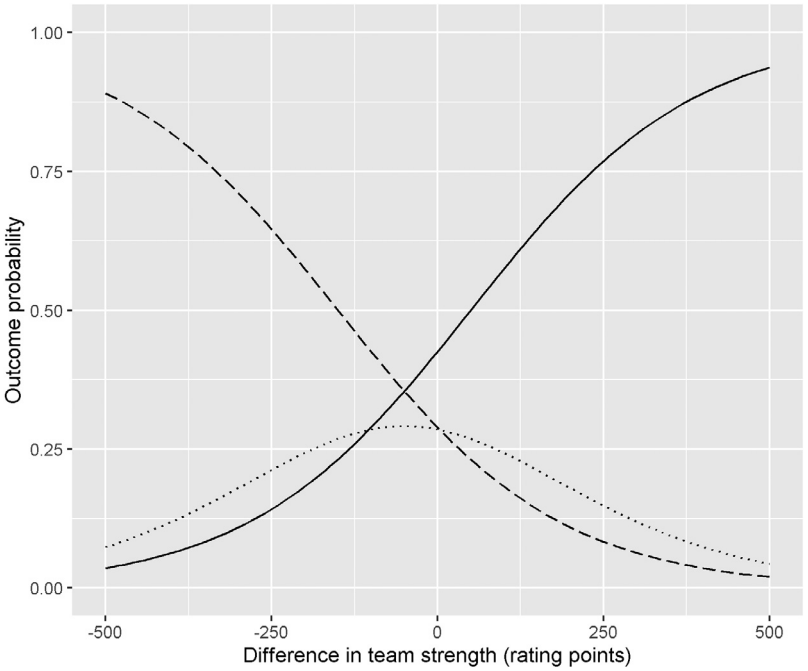


Fig. 1. Probability of home wins (solid line), draws (dotted line), and away wins (dashed line) as functions of the difference in team ratings Δ. Probabilities are based on a model with parameters $c_0 = -0.9$, $c_1 = 0.3$, and $\beta = 0.006$.

fully consistent with the competition format of major European football leagues such as the English Premier League, Spanish La Liga, Italian Serie A, and French Ligue 1) in 10 consecutive seasons. For each match, the ground truth (true rating for each team and point in time; true forecast for each match) is simulated, as well as the rating and probability estimates (i.e., forecasts) by bookmaker and bettor. True ratings of the teams are simulated as time-dependent processes with random fluctuations and random seasonal trends. True forecasts are simulated based on OLR, where we are free in choosing the parameters, but favor a realistic reflection of real-world football. We choose $c_0 = -0.9$, $c_1 = 0.3$ and $\beta = 0.006$ as the true model, which is influenced by previous experience of fitting OLR models on real-world football data, seems to be highly consistent with the probability distribution illustrated by Hvattum and Arntzen (2010) and has some favorable properties like round parameter values and an implicit home advantage (see Nevill & Holder, 1999; Wunderlich, Weigelt, Rein, & Memmert, 2021) of 50 rating points, which means that home and away winning probability are equal if the away teams' rating exceeds the home teams' rating by 50 points. All simulations are based on a simulation framework capable of capturing all the above-mentioned elements (Garnica-Caparrós et al., 2022) and we refer to this paper for more details on the framework and the advantages of artificial data.

4.2. Biases

In knowledge of the true probability model, specific biases of the bookmaker and the forecaster can be introduced in two ways. First, the true ratings might be misestimated and thus the input of the probability model. In the experiments, we assume that both bookmaker and forecaster have unbiased, but slightly inaccurate estimates of the true ratings. This is realized by adding a normally distributed error term to the true ratings ($\epsilon_{\text{Bookie}} \sim N(0, 25)$ for the bookmaker model and $\epsilon_{\text{Bettor}} \sim N(0, 50)$ for the forecaster model). Second, the probability model itself might be an inaccurate reflection of the true model. Bookmaker and forecaster are assumed to utilize the same family of models, which restricts biases to inaccurate model parameters. For the bookmaker we choose model parameters $c_0^* = -0.8$, $c_1^* = 0.4$, $\beta^* = 0.004$, leading to overestimation (underestimation) of the outsider (favorite), the so-called favorite longshot bias (Hegarty & Whelan, 2024; Ottaviani & Sorensen, 2008). On the forecaster side, we assume that a variety of model specifications are used for optimization, where all these specifications (apart from the one true model) will involve some type of systematic bias.

4.3. Results

Table 3 summarizes the results from artificial data. Results are based on optimization across 144 forecaster models (i.e., model specifications) representing all combinations of the parameters c_0 (ranging from –1.2 to –0.7 with steps of 0.1), c_1 (from 0.0 to 0.5 with steps of 0.1), and β (from 0.002 to 0.008 with steps of 0.002). Parameters and steps were chosen with the aim of including the true model and relevant biased models, representing a reasonable granularity, and still ensuring an acceptable running time of the simulations. In the accuracy-optimized case, the chosen model specification leads to

Table 2
Framework with its important concepts to compare accuracy-optimized with profitability-optimized models on both artificial and real-world data.

Concept	Explanation	Artificial data	Real-world data
Ground Truth			
True Rating	The inherent strength of a football team quantified as a single number	R_T simulated as time-dependent process (see Garnica-Caparrós, Memmert, & Wunderlich, 2022)	Unknown
True Forecast (True Model)	The process translating team strengths into outcome probabilities	Simulated based on OLR model (see Hvattum & Arntzen, 2010)	Unknown
Bookmaker			
Rating Bookmaker	The implicit estimation of a rating (i.e. team strength) by the bookmaker	$R_T + \epsilon_{Bookie} \sim N(0, 25)$	Unknown
Bias Bookmaker	A specific bias by the bookmaker reflected in the forecast and implying suboptimal forecasts	Favorite longshot bias deliberately introduced to the bookmaker model	Exact biases unknown, but e.g. evidence on favorite longshot bias (Hegarty & Whelan, 2024)
Forecast Bookmaker	The estimation of outcome probabilities by the bookmaker (that might imply a systematic bias)	Obtained using a biased OLR model ($c_0^* = -0.8$, $c_1^* = 0.4$, $\beta^* = 0.004$) on the bookmaker ratings	Implicitly contained in the betting odds (market prices)
Bettor			
Rating Bettor	The estimation of a rating (i.e. team strength) by the bettor	$R_T + \epsilon_{Bettor} \sim N(0, 50)$	ELO Rating, k = 20 (see Hvattum & Arntzen, 2010)
Forecast Bettor	The estimation of outcome probabilities by the bettor	144 OLR models with parameters c_0 (-1.2 to -0.7, steps of 0.1), c_1 (0.0 to 0.5, steps of 0.1) and β (0.002 to 0.008, steps of 0.002) to be optimized	125 OLR models with parameters c_0 (-1.2 to -0.8, steps of 0.1), c_1 (0.0 to 0.4, steps of 0.1) and β (0.004 to 0.0012, steps of 0.002) to be optimized
General aspects			
N	The number of observations (i.e. matches)	152,000	65,457
Margin	The margin contained in the betting odds leading to a bookmaker advantage	10%	Implicitly contained in the betting odds (Average odds: 6.5%, Maximum odds: 0.9%)
Data Split	Split of data into in-sample and out-of-sample data	By time, 5 seasons in sample ($N = 76{,}000$), 5 seasons out of sample ($N = 76{,}000$)	By time, 1 season rating initialization ($N = 3{,}509$), 9 seasons in sample ($N = 30{,}692$), 9 seasons out of sample ($N = 31{,}256$)
Statistical Measure	The statistical measure used to validate the accuracy of forecasts	Likelihood	
Economic Measure	The economic measures used to validate the profitability of forecasts	Actual Betting Returns	Actual Betting Returns and Expected (i.e., noise-free) Betting Returns

small losses in the observed betting returns. When profitability optimization is applied, a different model specification is selected and the betting returns are slightly positive, which indicates that the profitability-optimized model outperforms the accuracy-optimized model. We refrain from using any conventional statistical test to check the statistical significance of this difference. Instead, artificial data offer a convenient alternative method to exclude random influences from the results. As the true probability of each outcome is known in the generated data, we can infer the expected (i.e., noise-free) betting returns based on the betting decisions and these probabilities. Any difference in expected returns represents a true difference and not one caused by statistical uncertainty in the out-of-sample data. Driven by the large number of matches, the expected betting returns are roughly consistent with the observed betting returns and, most importantly, confirm the existence of a substantial difference between the models. In order to exclude the possibility that differences are an artefact of the number of bets made, we additionally report returns per bet, where the overperformance of profitability optimization still holds true.

4.4. Analysis

Model parameters of the OLR are difficult to grasp intuitively; thus we illustrate the probabilities implied by three relevant models (the true model, the biased bookmaker model, and the profitability-optimized model) in Fig. 2.

The bookmaker model, as intended in the simulation, represents a favorite longshot bias, where the winning probabilities of favorites (outsiders) are underestimated (overestimated) in relation to the true model. Moreover, it is apparent that—in relation to the true model—the profitability-optimized model is biased in the opposite direction than the bookmaker. This is a direct consequence of the fact that the direction of forecasting error influences profitability, which has been evidenced before in

Table 3
Model selection and model performance when using accuracy optimization or profitability optimization on artificial data.

Method	Accuracy optimization			Profitability optimization			Difference
Measure for optimization (IS)	Likelihood			Betting Returns			
Model Selection (IS)	c_0	c_1	β	c_0	c_1	β	
	−0.8	0.3	0.006	−1.2	0.0	0.008	
Number of Bets	48 159			53 354			
Observed Betting Returns (OOS)	−394.48			463.41			857.89
Observed Betting Returns per Bet	-.0082			.0087			.0169
Expected Betting Returns (OOS)	−463.66			554.26			1017.92
Expected Betting Returns per Bet	−0.0096			.0104			.0200

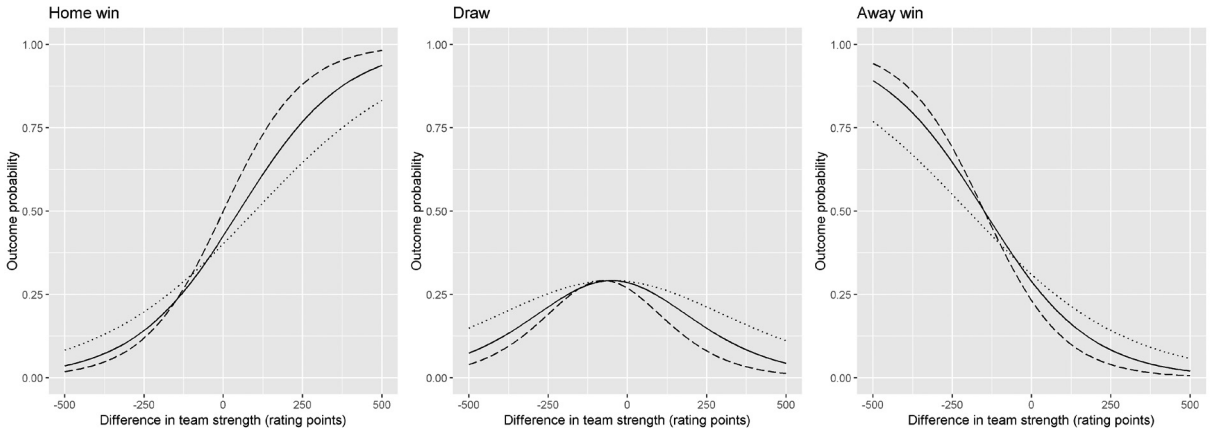


Fig. 2. Visualization of OLR forecasting models. The *x*-axis refers to the rating difference in points as a measure of relative team strength and the *y*-axis refers to the probability of a match outcome implied by the model. Solid lines refer to the true forecasting model, while dotted lines refer to the favorite longshot biased bookmaker model and dashed lines refer to the profitability-optimized model.

sports (Wunderlich & Memmert, 2020) and other domains (Hobbs et al., 1999). Our experiment implies that profitability optimization in fact enables selection of models opposing the market error and thus specifically exploiting a bookmaker bias. Accuracy optimization, while being more successful in estimating the truth (true model), is not designed to do that and also misses important information (the market prices) to do so. Please note that the accuracy-based model slightly misestimates the true model used to generate the artificial data. It is not shown in the figure, as it is very close to the true model and in theory, for an infinitely large number of observations, should equal the true model.

4.5. Robustness tests

To make sure that results are not contingent upon our design choices in the data generation, we employ several robustness tests. First, we additionally analyze a theoretically margin-free market (bookmaker margin of 0%). Second, we replace the fixed betting strategy of always betting one unit by a non-multiplicative Kelly betting strategy that assumes a stake of $\frac{o_i p_i - 1}{o_i - 1}$ for each profitable bet, where o_i are the odds and p_i is the forecast probability for outcome i (see Hvattum & Arntzen, 2010). Third, we test two additional bookmaker biases, namely a model systematically overestimating the home advantage (represented by OLR with $c_0^* = -1.2$, $c_1^* = 0.0$, $\beta^* = 0.006$) and a model systematically underestimating the occurrence of draws (represented by OLR with $c_0^* = -0.8$, $c_1^* = 0.2$, $\beta^* = 0.006$). Results are summarized in Table 4. For reasons of simplicity, we content ourselves with reporting only expected (i.e., randomness-free) returns and only differences between profitability and accuracy optimization. The main result, profitability optimization leading to higher absolute returns out of sample, is strongly supported by the robustness tests, where all absolute differences of returns and all differences of returns per bet are positive.

5. Evidence from real-world data

5.1. Dataset

The real-world dataset includes 65,457 matches from 11 domestic football leagues (first divisions of Belgium, England, France, Germany, Greece, Italy, the Netherlands, Portugal, Scotland, Spain, and Turkey) from 19 seasons (2005/06 to 2023/24). For each match, it covers data on the league, the date of the match, and the final result, as well as averaged betting odds and maximum betting odds for the outcomes home, draw, away. Data were obtained from www.football-data.co.uk, a well-known and widely used resource in the forecasting literature We excluded matches with missing data on the betting odds (38 matches). Moreover, as the absence of data errors can be highly crucial to the results of forecasting studies (see Clegg & Cartlidge, 2024; Ramirez, Reade, & Singleton,

Table 4
Robustness tests for comparing accuracy optimization and profitability optimization on artificial data. The table summarizes the out-of-sample profitability of models chosen when optimizing for either accuracy or profitability in sample. Absolute and relative differences between the two methods are reported for different biases, betting strategies and margins, where the first row equals the results of Table 3.

Bias	Betting strategy	Margin	Differences	
			Exp. Returns absolute	*Exp. Returns per bet*
Favorite Longshot	Fixed	10%	**1 017.92**	**.0200**
Favorite Longshot	Fixed	0%	**1 340.21**	**.0183**
Favorite Longshot	Kelly	10%	**465.05**	**.0078**
Favorite Longshot	Kelly	0%	**1 857.98**	**.0236**
Home Advantage	Fixed	10%	**1 390.94**	**.0033**
Home Advantage	Fixed	0%	**2136.18**	**.0062**
Home Advantage	Kelly	10%	**426.86**	**.0016**
Home Advantage	Kelly	0%	**783.45**	**.0009**
Draw	Fixed	10%	**3 994.40**	**.0530**
Draw	Fixed	0%	**2 574.97**	**.0253**
Draw	Kelly	10%	**1 087.56**	**.0129**
Draw	Kelly	0%	**2 682.74**	**.0249**

2023), sanity checks were performed. We checked the absence of doubled matches in the data, the correctness of atypical results with more than 10 goals scored by one team, and whether the results (home, draw, away) were consistent with the goals scored by each team. All data were correct in this regard, but we identified and deleted one match from the data that had been cancelled and awarded by the football association. Moreover, we decided to exclude matches where average odds were higher than maximum odds (1 match), the sum of inverse average odds fell below 100% or exceeded 120% (0 matches), or the sum of inverse maximum odds fell below 95% or exceeded 115% (81 matches). This limited the original dataset of 65,578 matches to 65,457 matches. In contrast to the artificial data, the ground truth for ratings and forecasts is not directly observable in real-world data and the exact process of the bookmaker coming up with betting odds (including any potential biases) is unknown. To estimate the unknown ratings, we use an ELO rating with an adjustment parameter of $k = 20$, as proposed by Hvattum and Arntzen (2010).

5.2. Biases

In real-world data, biases cannot be added deliberately and, in most cases, are difficult to detect directly from the betting odds data. Yet our analysis is dependent on the existence of biases, as in a perfectly unbiased market, every bet should have the same expected value and lead to an equal level of profitability. A bias that is well studied and easily observable in real-world data without any modeling assumptions is the favorite longshot bias, which has been studied for a long time and has recently been reported again by Hegarty and Whelan (2024). Fig. 3 uses the present dataset to replicate Fig. 2 of the aforementioned study, yet illustrating the bias for both average and maximum odds. While successfully replicating the results for the average odds, we show that the bias is significantly reduced for maximum odds, but appears to be present to a smaller extent. What is important in the context of the current study is that the favorite longshot bias can be captured by the specification of an OLR model. Please note that this is not the case for more nuanced market inefficiencies, that is, on team level (Braun & Kvasnicka, 2013) or based on specific events (Bernardo et al., 2019), which are not traceable without additional information and modeling effort and thus are not well captured by OLR.

5.3. Results

Table 5 summarizes the results based on real-world data. Results are based on optimization across 125 forecaster models (i.e., model specifications) representing all combinations of the parameters c_0 (ranging from –1.2 to –0.8 with steps of 0.1), c_1 (from 0.0 to 0.4 with steps of 0.1), and β (from 0.004 to 0.012 with steps of 0.002), which were chosen to ensure a reasonable range of parameters around the typical parameters resulting from accuracy optimization in this domain. Consistent with the artificial data case, different model specifications are selected and the profitability-optimized model outperforms the accuracy-optimized model both for absolute betting returns and for returns per bet. In real-world data, we cannot infer expected betting returns and thus need statistical methods to quantify randomness in the observed betting returns. Bootstrapping methods are used to generate resamples from the betting returns of both models and to calculate the difference between profitability- and accuracy-optimized model based on a total of 10,000 resamples. The 95% confidence interval for the difference in betting returns of 969.34 is (226.31, 1708.27), where the prevalence of differences smaller or equal to 0 is 0.55% (we will denote this value as "Bootstrap Probability $\leq$0" subsequently). We can therefore infer with a sufficiently low degree of insecurity that differences are actually systematic and not based on random fluctuations in the returns.

5.4. Analysis

The analysis for artificial data cannot be repeated equally for real-world data, as we know neither the true model, nor the biased bookmaker model. However, we can illustrate the OLR model specification best representing the true outcomes, that is, simply the accuracy-optimized model. Fig. 4 compares this model with the

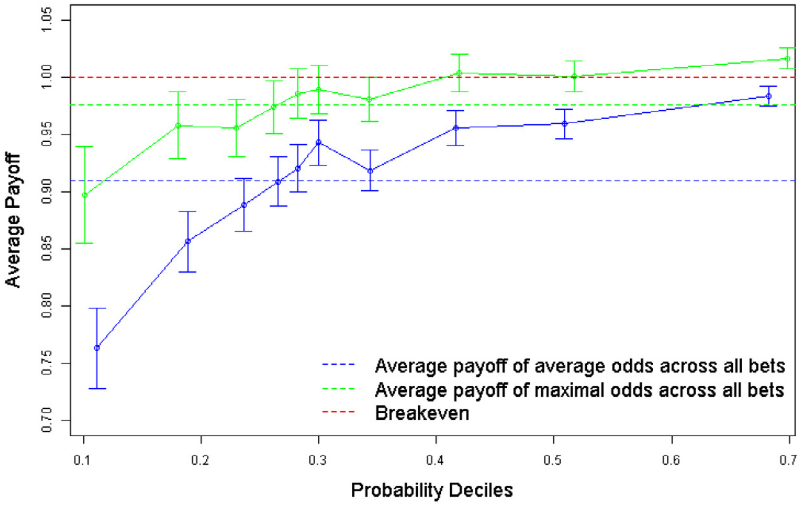


Fig. 3. Illustration of the favorite longshot bias in the betting market. The figure is largely adopted from the work of Hegarty and Whelan (2024), but based on the present dataset. The *x*-axis refers to the average probabilities implied in betting odds and the *y*-axis to average payoffs when betting a unit of one on each possible outcome. Each blue dot refers to a decile of the probabilities and error bars represent 95% confidence intervals. Results based on average betting odds are illustrated in blue and results based on maximal betting odds are illustrated in green.

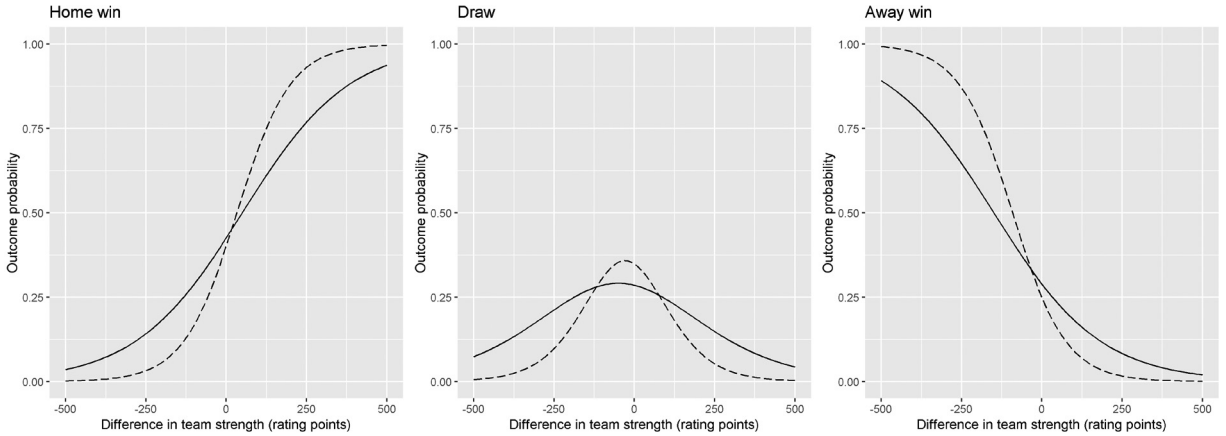


Fig. 4. Visualization of OLR forecasting models. The *x*-axis refers to the rating difference in points as a measure of relative team strength and the *y*-axis refers to the probability of a match outcome implied by the model. Solid lines refer to the accuracy-optimized forecasting model, while dashed lines refer to the profitability-optimized model.

Table 5
Model selection and model performance when using accuracy-optimization or profitability-optimization on real data.

Method	Accuracy optimization			Profitability optimization			Difference
Measure for optimization (IS)	Likelihood			Betting Returns			
Model Selection (IS)	c_0	c_1	β	c_0	c_1	β	
	−0.9	0.3	0.006	−1.1	0.4	0.012	
Number of Bets	25,480			36,741			
Observed Betting Returns (OOS)	−3640.98			−2671.64			969.34
Observed Betting Returns per Bet	−.1429			−.0727			.0702

Table 6
Robustness tests for comparing accuracy optimization and profitability optimization on real-world data. The table summarizes the out-of-sample profitability of models chosen when optimizing for either accuracy or profitability in sample. Absolute and relative differences between both methods as well as the prevalence of differences equal or lower than 0 based on bootstrapping are reported for different odds types and betting strategies, where the first row equals the results of Table 5.

Odds type	Betting strategy	Differences		
		Returns absolute	*Bootstrap probability* ≤ 0	*Returns per bet*
Average	Fixed	**969.34**	**.0055**	**.0702**
Average	Threshold (20%)	**410.74**	**.0394**	**.1097**
Average	Kelly	**−23.30**	**.7891**	**–.0016**
Maximum	Fixed	**816.88**	**.0484**	**.0201**
Maximum	Threshold (30%)	**243.71**	**.2033**	**.0297**
Maximum	Kelly	**23.52**	**.3352**	**.0007**

profitability-optimized model. In analogy to the artificial data, we see that the profitability-optimized model counters a favorite longshot bias, which, as we know from Fig. 3, exists in the real-world betting market.

5.5. Robustness tests

In real-world data, opportunities for robustness tests are limited, as we cannot deliberately change margins or introduce biases into the data. However, we can repeat analysis using maximum instead of average odds and use different betting strategies. Apart from the Fixed and Kelly betting strategies already used in the artificial data, we employ an additional strategy in an attempt to identify profitable bets. For a given threshold, the strategy denoted as Threshold bets one unit if $\tilde{p}_i \cdot o_i - 1 >$ threshold and does not bet otherwise, thus considering the amount of win expected by a bet under the assumption that the forecasted probability is correct. Table 6 shows results of these robustness tests, reporting the observed absolute returns and returns per bet as well as the bootstrap probability of differences smaller or equal to zero. Results for the robustness tests are more ambiguous than for the artificial data, which is presumably explainable—at least in part—by the fact that biases in real-world data are more nuanced and thus opportunities for profitable bets more limited if compared to artificial data. However, we still find that profitability optimization leads to higher absolute (and relative) returns in five of six cases. For a Fixed strategy with both odds types and a Threshold strategy for average odds, out-of-sample randomness in the betting returns as a reason for this result can be excluded at a level of 5%. The Kelly strategies show the least clear results, with returns in Kelly Average even falling short of the accuracy-optimized model. Please note that all out-of-sample returns, even for the Threshold models, were negative (not displayed in the Table), which implies that profitability optimization, although improving betting returns in most cases, was not enough to reach profitability. Thus, the usage of a simple Elo model, consistent with previous results (Hvattum & Arntzen, 2010), is not sufficient to beat the market.

6. Discussion

The present analysis was motivated by the established idea of evaluating economic value in addition to forecasting accuracy (Hong et al., 2020; Koopman & Lit, 2015; Murphy, 1993). In the literature, it is well evidenced that the two goals are not necessarily in line (Fuertes et al., 2015; Hubáček & Šír, 2023; Yang & Kleissl, 2022), and particularly, besides accuracy, the direction of forecasting error may determine decision consequences, including economic value (Hobbs et al., 1999; Leitch & Tanner, 1991; Wunderlich & Memmert, 2020). Consequently, this paper argues that when aiming for model profitability, market prices should already be considered in model selection. The present analysis supports this idea by theoretical evidence as well as evidence from artificial data and real-world data on increased performance of profitability-optimized models compared with accuracy-optimized models. While this result sounds remarkably intuitive, it is not self-evident, as it has been shown that under specific circumstances statistical optimization of accuracy likewise optimizes economic value of models (Johnstone, 2011). Moreover, this topic has not yet been discussed sufficiently, at least in the domain of sports forecasting, where it is common to report out-of-sample betting returns, while relying fully on accuracy measures for optimizing the models in sample (Boshnakov et al., 2017; Hvattum & Arntzen, 2010; Koopman & Lit, 2015, 2019). To the best of our knowledge, the only exception considering market prices in model selection is the work of Hubáček et al. (2019) using decorrelation of market and forecast in model training. Interestingly, Grant and Johnstone (2010), besides accuracy measures, have used a Kelly score based on in-sample Kelly betting returns in an attempt to identify effective forecasters (not models). While the results were more promising than for conventional accuracy measures, this aspect was not analyzed further.

It is important to note that the idea and methods used in this study are independent of the specific type of market bias. While studies on market efficiency usually assume a certain type of inefficiency pre-analysis (Bernardo et al., 2019; Braun & Kvasnicka, 2013; Direr, 2011; Meier, Flepp, & Franck, 2021), no assumption on the type of bias needs to be made before profitability optimization. As long as the model family is able to capture the relevant type of biases, the profitability optimized model should be able to exploit the market bias implicitly. This idea is backed up by our analyses, as the models are able to detect and counter several biases without any explicit consideration of these biases in the model or optimization process. In real-world applications, this is a nice property, as it theoretically bears the possibility of

exploiting types of biases that the forecaster is not even aware of.

It is important to distinguish clearly what the idea and characteristics of accuracy- and profitability-optimized models in the context of this study are and what they are not. Profitability optimization of models can be considered a pre-forecast and not post-forecast economic decision, as the idea is to adjust the forecast itself based on in-sample profitability and not to optimize the economic value of decisions based on the very same forecast(s) (Zhu et al., 2002). Moreover, accuracy-optimized models in this study are assumed a priori to be non-perfect forecasts (i.e., not fully reflecting the true probabilities). In previous literature, the economic value of forecasts has been tested in comparison with the benchmark of a perfect forecast (Lee & Lee, 2007; Zareipour et al., 2009). This is not the case in our analyses, where we deliberately introduce inaccuracies in estimating the team strengths for bookmaker and forecaster into the artificial data and expect that such inaccuracies can be existent in the real-world data. Thus, even when selecting the true probability model, the forecaster will not obtain perfectly accurate probabilities and we do not conclude that profitability-optimized forecasts are superior to perfect forecasts, but that they are superior to a priori imperfect accuracy-optimized forecasts.

Obviously, the present analysis is subject to some limitations, mainly attributable to the assumptions taken in theoretical considerations, artificial data generation and general model assumptions. The clarity and intensity of bookmaker biases introduced in theory and artificial data is a simplification of real-world betting markets. In real-world data, several biases can exist concurrently, but presumably they are rather nuanced, if existent at all, which limits the bettors' possibilities of identifying profitable bets. Moreover, betting odds can be affected by numerous additional aspects such as book balancing to manage risk (Stark & Cortis, 2017), effects of different market structures (Franck, Verbeek, & Nüesch, 2010), structural differences in bookmaker margins (Štrumbelj, 2014), or aspects of marketing and reaction to competitor prices. Finally, some of the potential real-world biases have a different structure than the ones considered in this study, as they might depend on team-specific information (Forrest & Simmons, 2008) or non-match-related events (Bernardo et al., 2019). We assume the structure and intensity of biases to be important reasons that results in the real-world data are far more ambiguous than those in artificial data. Inclusion of further biases with a different structure, however, would require additional modeling effort (artificial data) or more granular information (real-world data), and thus remain as relevant aspects to be tackled in future work.

Another limitation is that we restricted analysis to OLR models. In artificial data this model family equals the model family of the true model and biased bookmaker model, which favors the possibility of constructing profitable models. In the real-world data, however, the implicit true model will be more complicated and (as argued before) include more information than possibly reflected in Elo ratings and OLR models. In fact, profitability optimization improves betting returns on real-world data, but still achieves negative profits. This is consistent with previously reported negative returns for OLR models (Hvattum & Arntzen, 2010) and might be explained by the rather simple model structure and limited usage of information. More complex model families were not considered within the current study, as this would require a more granular real-world dataset and additional modeling effort in the artificial data generation and (in the case of machine learning models) would reduce the interpretability of results.

Future research could investigate more complex models (both statistical model families and machine learning models) and analyze the impact of the model family on the results. Moreover, it would be valuable to investigate whether the present approach can be successfully transferred to real-world data from other domains of forecasting. Given that our approach of model selection was a very simple grid search testing a predefined range of models and picking the best-performing one, we would encourage discussion on improved methods for profitability optimization. The fact that established statistical methods such as maximum likelihood estimation are not specifically designed to consider market prices might be one reason for the underconsideration of profitability optimization in the literature so far. Improved methods might be either specific methods to optimize profitability of model parameters or inclusion of market prices in traditional accuracy-based methods like regression models. Another interesting approach worth investigating could be the effect of measures reflecting accuracy and profitability concurrently for model selection on the out-of-sample model profitability.

7. Conclusions

The present study has tackled the problem of optimal decision making when selecting promising forecasting models in terms of profitability. Based on theoretical calculations, we show that the consistency of accuracy-optimized and profitability-optimized models that had been evidenced in earlier research is contingent upon very specific circumstances and does not hold when strict assumptions are relaxed. Based on artificial datasets, it is demonstrated that superiority of profitability optimization (through consideration of market prices in model selection) compared with accuracy optimization of models also holds in more realistic settings. Through profitability optimization in our approach, the forecaster was able to identify efficient models exploiting three different bookmaker biases without requiring a priori specification of the bias expected in the market. We find the same result, albeit less clear and more nuanced, in real-world data, which is evidence for the transferability of our results to real-world applications in the domain of sports betting markets. In future work, more discussion on the role of profitability optimization is desirable as well as development and analysis of improved method for profitability optimization of forecasting models.

CRediT authorship contribution statement

Fabian Wunderlich: Writing – original draft, Visualization, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Marc Garnica Caparrós:** Writing – review & editing, Validation, Software, Methodology, Data curation, Conceptualization. **Daniel Memmert:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization.

Funding

This research was partially supported by the German Research Foundation (DFG, Deutsche Forschungsgemeinschaft) under Grant ME 2678/29-3 | WU 1005/1-3 to D.M. and F.W. The simulation framework serving as a basis for the artificial data generation was supported by the German Research Foundation (DFG, Deutsche Forschungsgemeinschaft) in an earlier Grant (ME 2678/29.1 to D.M.).

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used DeepL to improve English language and ChatGPT to assist in coding to create graphical illustrations. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We would like to thank Lukas Gardeweg for his support in the visualization of results.

Data and code availability statement

All source code used for the present analyses is made public as an online repository and can be accessed at https://github.com/spoho-datascience/market-model-profitability. In case of real-world data, all relevant data sources are publicly available and cited in the manuscript. In case of artificial datasets, these were generated and can be regenerated by simulation using the code in the repository. In case of non-deterministic results (artificial data generation, bootstrapping), the random number generators in the code use a seed ensuring reproducibility of the exact data and results reported in the manuscript.

References

Angelini, G., & Angelis, L. de (2019). Efficiency of online football betting markets. *International Journal of Forecasting*, *35*(2), 712–721. http://dx.doi.org/10.1016/j.ijforecast.2018.07.008.

Bernardo, G., Ruberti, M., & Verona, R. (2019). Semi-strong inefficiency in the fixed odds betting market: Underestimating the positive impact of head coach replacement in the main European soccer leagues. *The Quarterly Review of Economics and Finance*, *71*, 239–246. http://dx.doi.org/10.1016/j.qref.2018.08.007.

Boothe, P., & Glassman, D. (1987). Comparing exchange rate forecasting models. *International Journal of Forecasting*, *3*(1), 65–79. http://dx.doi.org/10.1016/0169-2070(87)90079-3.

Boshnakov, G., Kharrat, T., & McHale, I. G. (2017). A bivariate Weibull count model for forecasting association football scores. *International Journal of Forecasting*, *33*(2), 458–466. http://dx.doi.org/10.1016/j.ijforecast.2016.11.006.

Braun, S., & Kvasnicka, M. (2013). National sentiment and economic behavior. *Journal of Sports Economics*, *14*(1), 45–64. http://dx.doi.org/10.1177/1527002511414718.

Cerdá Tena, E., & Quiroga Gómez, S. (2011). Economic value of weather forecasting: the role of risk aversion. *TOP*, *19*(1), 130–149. http://dx.doi.org/10.1007/s11750-009-0114-3.

Clegg, L., & Cartlidge, J. (2024). Not feeling the buzz: Correction study of mispricing and inefficiency in online sportsbooks. *International Journal of Forecasting*, http://dx.doi.org/10.1016/j.ijforecast.2024.06.012, Advance online publication.

Constantinou, A. C., Fenton, N. E., & Neil, M. (2012). pi-football: A Bayesian network model for forecasting Association Football match outcomes. *Knowledge-Based Systems*, *36*, 322–339. http://dx.doi.org/10.1016/j.knosys.2012.07.008.

Direr, A. (2011). Are betting markets efficient? Evidence from European Football Championships. *Applied Economics*, *45*(3), 343–356. http://dx.doi.org/10.1080/00036846.2011.602010.

Ertimur, Y., Sunder, J., & Sunder, S. V. (2007). Measure for measure: The relation between forecast accuracy and recommendation profitability of analysts. *Journal of Accounting Research*, *45*(3), 567–606. http://dx.doi.org/10.1111/j.1475-679X.2007.00244.x.

Fama, E. F. (1970). Efficient capital markets: A review of theory and empirical work. *The Journal of Finance*, *25*(2), 383–417. http://dx.doi.org/10.2307/2325486.

Forrest, D., & Simmons, R. (2008). Sentiment in the betting market on Spanish football. *Applied Economics*, *40*(1), 119–126. http://dx.doi.org/10.1080/00036840701522895.

Franck, E., Verbeek, E., & Nüesch, S. (2010). Prediction accuracy of different market structures — bookmakers versus a betting exchange. *International Journal of Forecasting*, *26*(3), 448–459. http://dx.doi.org/10.1016/j.ijforecast.2010.01.004.

Fuertes, A.-M., Kalotychou, E., & Todorovic, N. (2015). Daily volume, intraday and overnight returns for volatility prediction: profitability or accuracy? *Review of Quantitative Finance and Accounting*, *45*(2), 251–278. http://dx.doi.org/10.1007/s11156-014-0436-6.

Garnica-Caparrós, M., Memmert, D., & Wunderlich, F. (2022). Artificial data in sports forecasting: a simulation framework for analysing predictive models in sports. *Information Systems and E-Business Management*, http://dx.doi.org/10.1007/s10257-022-00560-9, Advance online publication.

Gneiting, T. (2011). Making and evaluating point forecasts. *Journal of the American Statistical Association*, *106*(494), 746–762. http://dx.doi.org/10.1198/jasa.2011.r10138.

Grant, A., & Johnstone, D. (2010). Finding profitable forecast combinations using probability scoring rules. *International Journal of Forecasting*, *26*(3), 498–510. http://dx.doi.org/10.1016/j.ijforecast.2010.01.002.

Hegarty, T., & Whelan, K. (2024). Forecasting soccer matches with betting odds: A tale of two markets. *International Journal of Forecasting*.

Hobbs, B. F., Jitprapaikulsarn, S., Konda, S., Chankong, V., Loparo, K. A., & Maratukulam, D. J. (1999). Analysis of the value for unit commitment of improved load forecasts. *IEEE Transactions on Power Systems*, *14*(4), 1342–1348.

Hong, T., Pinson, P., Wang, Y., Weron, R., Yang, D., & Zareipour, H. (2020). Energy forecasting: A review and outlook. *IEEE Open Access Journal of Power and Energy*, *7*, 376–388. http://dx.doi.org/10.1109/OAJPE.2020.3029979.

Hubáček, O., & Šír, G. (2023). Beating the market with a bad predictive model. *International Journal of Forecasting*, *39*(2), 691–719. http://dx.doi.org/10.1016/j.ijforecast.2022.02.001.

Hubáček, O., Šourek, G., & Železný, F. (2019). Exploiting sports-betting market using machine learning. *International Journal of Forecasting*, *35*(2), 783–796. http://dx.doi.org/10.1016/j.ijforecast.2019.01.001.

Hvattum, L. M., & Arntzen, H. (2010). Using ELO ratings for match result prediction in association football. *International Journal of Forecasting*, *26*(3), 460–470. http://dx.doi.org/10.1016/j.ijforecast.2009.10.002.

Johnstone, D. J. (2011). Economic interpretation of probabilities estimated by maximum likelihood or score. *Management Science*, *57*(2), 308–314. http://dx.doi.org/10.1287/mnsc.1100.1272.

Koopman, S. J., & Lit, R. (2015). A dynamic bivariate Poisson model for analysing and forecasting match results in the English Premier League. *Journal of the Royal Statistical Society: Series A (Statistics in Society)*, *178*(1), 167–186. http://dx.doi.org/10.1111/rssa.12042.

Koopman, S. J., & Lit, R. (2019). Forecasting football match results in national league competitions using score-driven time series models. *International Journal of Forecasting*, *35*(2), 797–809. http://dx.doi.org/10.1016/j.ijforecast.2018.10.011.

Lee, K.-K., & Lee, J.-W. (2007). The economic value of weather forecasts for decision-making problems in the profit/loss situation. *Meteorological Applications*, *14*(4), 455–463. http://dx.doi.org/10.1002/met.44.

Leitch, G., & Tanner, J. E. (1991). Economic forecast evaluation: profits versus the conventual error measures. *The American Economic Review*, 580–590.

Meier, P. F., Flepp, R., & Franck, E. (2021). Are sports betting markets semi-strong efficient? Evidence from the COVID-19 pandemic. *International Journal of Sport Finance*, *16*(3), http://dx.doi.org/10.32731/IJSF/163.082021.01.

Murphy, A. H. (1993). What is a good forecast? An essay on the nature of goodness in weather forecasting. *Weather and Forecasting*, *8*(2), 281–293.

Nevill, A. M., & Holder, R. L. (1999). Home advantage in sport. *Sports Medicine*, *28*(4), 221–236.

Ottaviani, M., & Sorensen, P. N. (2008). The favorite-longshot bias: An overview on the main explanations. In *Handbook of sports and lottery markets* (pp. 83–101). Elsevier.

Ramirez, P., Reade, J. J., & Singleton, C. (2023). Betting on a buzz: Mispricing and inefficiency in online sportsbooks. *International Journal of Forecasting*, *39*(3), 1413–1423. http://dx.doi.org/10.1016/j.ijforecast.2022.07.011.

Stark, D., & Cortis, D. (2017). Balancing the book: Is it necessary and sufficient? *The Journal of Gambling Business and Economics*, *11*(1), 1–6. http://dx.doi.org/10.5750/jgbe.v11i1.1280.

Štrumbelj, E. (2014). On determining probability forecasts from betting odds. *International Journal of Forecasting*, *30*(4), 934–943. http://dx.doi.org/10.1016/j.ijforecast.2014.02.008.

Wunderlich, F., & Memmert, D. (2020). Are betting returns a useful measure of accuracy in (sports) forecasting? *International Journal of Forecasting*, *36*(2), 713–722. http://dx.doi.org/10.1016/j.ijforecast.2019.08.009.

Wunderlich, F., & Memmert, D. (2021). Forecasting the outcomes of sports events: A review. *European Journal of Sport Science*, *21*(7), 944–957.

Wunderlich, F., Weigelt, M., Rein, R., & Memmert, D. (2021). How does spectator presence affect football? Home advantage remains in European top-class football matches played without spectators during the COVID-19 pandemic. *PLoS One*, *16*(3), Article e0248590. http://dx.doi.org/10.1371/journal.pone.0248590.

Yang, D., & Kleissl, J. (2022). Summarizing ensemble NWP forecasts for grid operators: Consistency, elicitability, and economic value. *International Journal of Forecasting*, http://dx.doi.org/10.1016/j.ijforecast.2022.08.002, Advance online publication.

Zareipour, H., Canizares, C. A., & Bhattacharya, K. (2009). Economic impact of electricity market price forecasting errors: A demand-side analysis. *IEEE Transactions on Power Systems*, *25*(1), 254–262.

Zhu, Y., Toth, Z., Wobus, R., Richardson, D., & Mylne, K. (2002). The economic value of ensemble-based weather forecasts. *Bulletin of the American Meteorological Society*, *83*(1), 73–83. http://dx.doi.org/10.1175/1520-0477(2002)083<0073:TEVOEB>2.3.CO;2.